Digital Logic Circuits (Spring 2026)

HW #3

14. Implement a logic circuit for the truth table in Table 5–9.

TABLE 5–9

Inputs				Output
A	***B***	***C***	***D***	***X***
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

14. $X = \overline{A}\,\overline{B}C\overline{D} + \overline{A}\,\overline{B}CD + \overline{A}B\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}D + A\overline{B}C\overline{D} + A\overline{B}CD + ABCD$
See Figure 5-8.

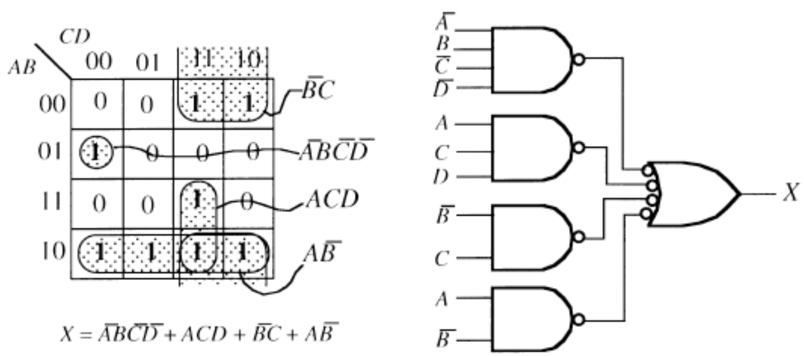


FIGURE 5-8

18. Minimize the gates required to implement the functions in each part of Problem 12 in SOP form.

(a) $X = \overline{A}B + CD + (\overline{A + B})(ACD + \overline{BE})$
(b) $X = ABC\overline{C}\overline{D} + D\overline{E}F + \overline{AF}$
(c) $X = \overline{A}[B + \overline{C}(D + E)]$

18. (a) $X = \overline{A}B + CD + (\overline{A+B})(ACD + \overline{B}\,\overline{E}) = \overline{A}B + CD + \overline{A}\,\overline{B}(ACD + \overline{B} + \overline{E})$

$= \overline{A}B + CD + \overline{A}\,\overline{B} + \overline{A}\,\overline{B}\,\overline{E} = \overline{A}(B + \overline{B}) + CD + \overline{A}\,\overline{B}\,\overline{E}$

$= \overline{A} + \overline{AB}\,\overline{E} + CD = \overline{A}(1 + \overline{BE}) + CD = \boldsymbol{\overline{A} + CD}$

See Figure 5-16.

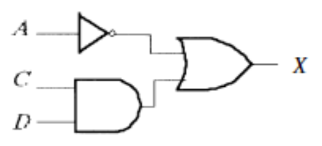


FIGURE 5-16

(b) $X = ABC\overline{C}\,\overline{D} + D\overline{E}F + \overline{A}\,\overline{F} = ABC\overline{C}\,\overline{D} + \overline{D}EF + \overline{A} + \overline{F}$

$= \boldsymbol{\overline{A} + B\overline{C}\,\overline{D} + \overline{F} + D\overline{E}}$

See Figure 5-17.

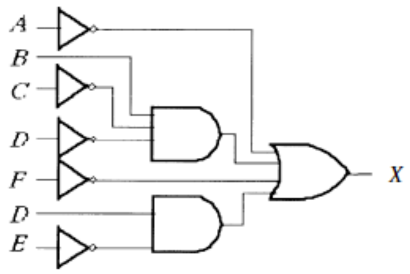


FIGURE 5-17

(c) $X = \overline{A}(B + \overline{C}(D + E)) = \overline{A}(B + \overline{C}D + \overline{C}E) = \boldsymbol{\overline{A}B + \overline{A}\,\overline{C}D + \overline{A}\,\overline{C}E}$

See Figure 5-18.

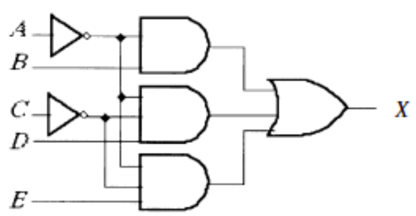
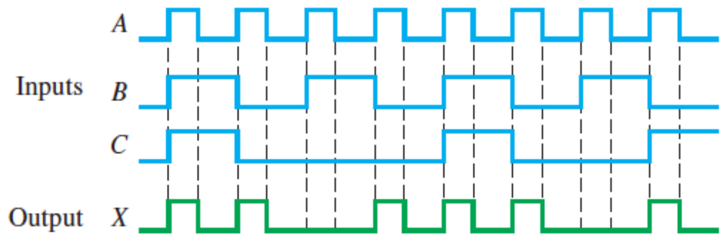


FIGURE 5-18

30. For the input waveforms in Figure 5–61, what logic circuit will generate the output waveform shown?



30. X is HIGH when ABC are all HIGH or when A is HIGH and B is LOW and C is LOW or when A is HIGH and B is LOW and C is HIGH.

$X = ABC + A\overline{B}\,\overline{C} + A\overline{B}C$

See Figure 5-53.

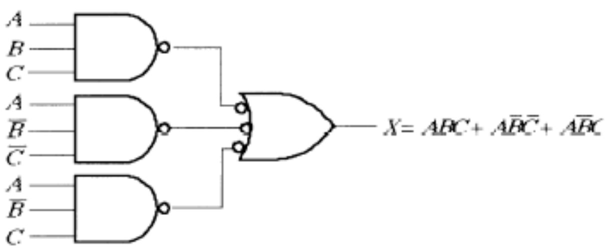
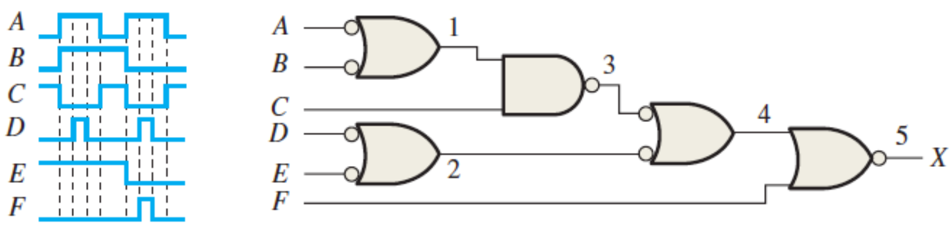


FIGURE 5-53

32. For the circuit in Figure 5–63, draw the waveforms at the numbered points in the proper relationship to each other.



32. See Figure 5-55.

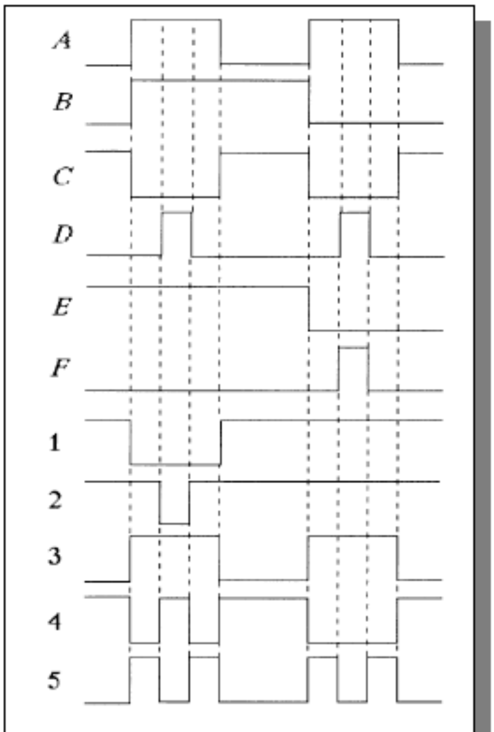


FIGURE 5-55

8. The input waveforms in Figure 6–72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.

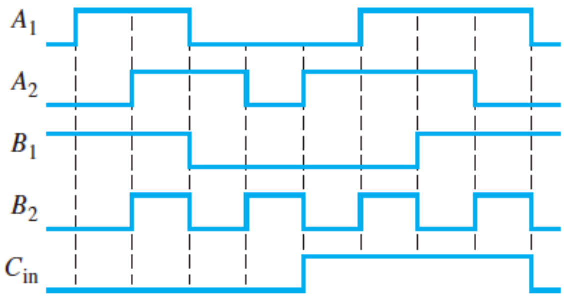


FIGURE 6–72

8. See Figure 6-3.

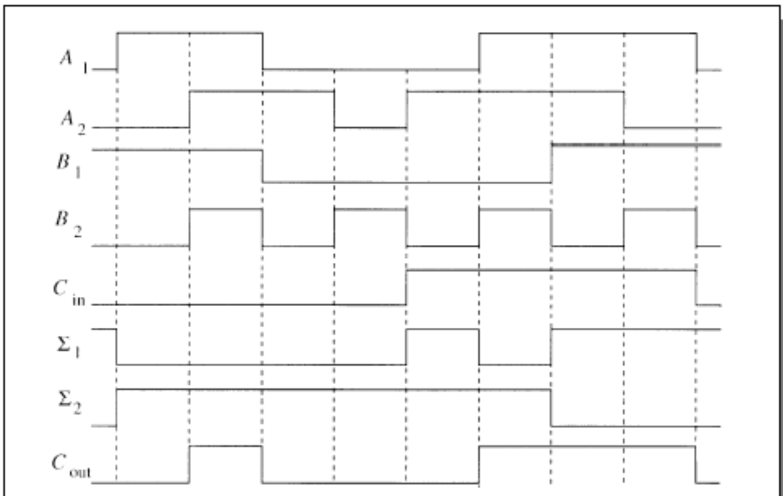


FIGURE 6-3

9. The following sequences of bits (right-most bit first) appear on the inputs to a 4-bit parallel adder. Determine the resulting sequence of bits on each sum output.

A_1	1010
A_2	1100
A_3	0101
A_4	1101
B_1	1001
B_2	1011
B_3	0000
B_4	0001

9.

A_4	A_3	A_2	A_1	B_4	B_3	B_2	B_1	Σ_5	Σ_4	Σ_3	Σ_2	Σ_1
1	0	0	1	0	0	0	1	0	1	0	1	0
1	0	1	0	1	1	0	1	0	0	1	1	1
0	0	1	0	0	0	1	1	0	0	1	0	1
1	0	1	1	0	1	1	1	1	0	0	1	0

$\Sigma_1 = 0110$
$\Sigma_2 = 1011$
$\Sigma_3 = 0110$
$\Sigma_4 = 0001$
$\Sigma_5 = 1000$

11. Each of the eight full-adders in an 8-bit parallel ripple carry adder exhibits the following propagation delay:

A to Σ and C_{out}:	20 ns
B to Σ and C_{out}:	20 ns
C_{in} to Σ:	30 ns
C_{in} to C_{out}:	25 ns

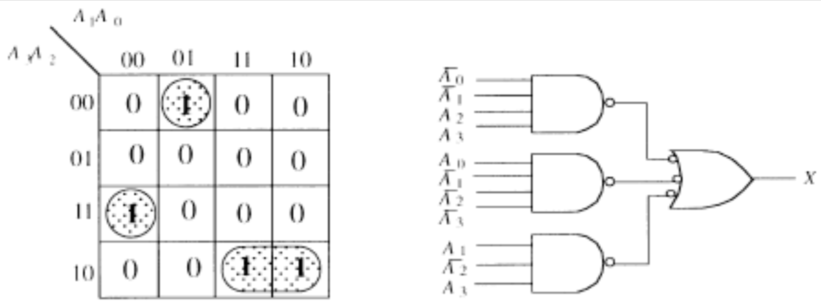
Determine the maximum total time for the addition of two 8-bit numbers.

11. $$t_{p(total)} = 30\,ns + 6 \times 25\,ns + 20\,ns = 200\,ns$$

19. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.

19. $X = \overline{A_3A_2A_1}A_0 + A_3\overline{A_2}A_1\overline{A_0} + A_3A_2\overline{A_1A_0} + A_3\overline{A_2}A_1A_0$

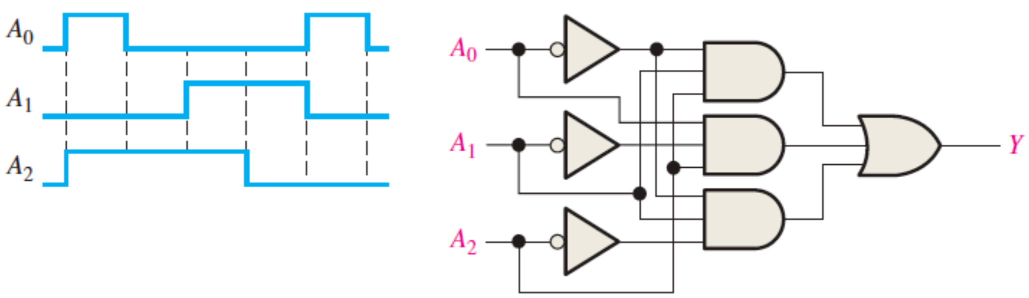
See Figure 6-9.



$X = \overline{A}_3\overline{A}_2\overline{A}_1A_0 + A_3A_2\overline{A}_1\overline{A}_0 + A_3\overline{A}_2A_1$

FIGURE 6-9

20. If the input waveforms are applied to the decoding logic as indicated in Figure 6–76, sketch the output waveform in proper relation to the inputs.



20. $Y = A_2 A_1 \overline{A_0} + A_2 \overline{A_1} A_0 + \overline{A_2} A_1 \overline{A_0}$

See Figure 6-10.

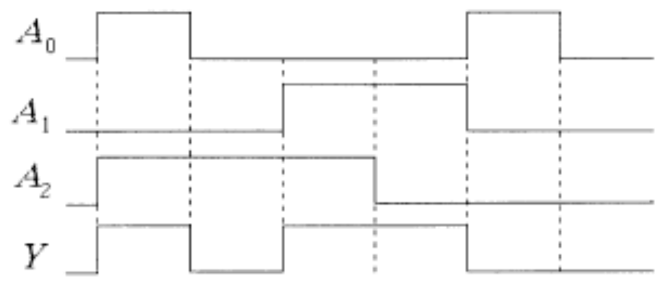


FIGURE 6-10